

Course Code: EET 305

Course Name: SIGNALS AND SYSTEMS

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

Marks

- 1 Define unit step signal. Sketch the signal $x(t) = -u(-t - 1)$ (3)
- 2 Check whether the system, $y(t) = e^{-2} x(t)$ is linear. (3)
- 3 Find the Fourier transform of $x(t) = e^{-t} u(t)$ (3)
- 4 Derive the elements in electrical system analogous to mechanical translational system with respect to force-voltage analogy. (3)
- 5 Determine the unit impulse response for the system with transfer function (3)
$$H(s) = \frac{10}{s+100}$$
- 6 List down any three properties of positive real functions. (3)
- 7 Define sampling and state sampling theorem. (3)
- 8 List down any three properties of z-transform. (3)
- 9 Using bilinear transformation method, check the stability of a discrete system (3)
having characteristic equation as: $z^2 + 0.2z + 0.1$.
- 10 Find the impulse response of a discrete time system having transfer function as: (3)
$$H(z) = \frac{z}{z-2}$$

PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

- 11 a) Differentiate between energy and power signals. Determine whether the signal, (7)
 $x(t) = e^{-t} u(t)$ is an energy signal or power signal.
- b) Find the convolution integral of the above signal $x(t) = e^{-t} u(t)$ with another (7)
signal $h(t) = e^{-3t} u(t)$

- 12 a) Test whether the following signals are periodic or not. If periodic, find the fundamental period (5)

$$x(t) = 5 \sin 24\pi t + 7 \sin 36\pi t$$

- b) With suitable examples, differentiate between (9)
- (a) Linear and non-linear systems
 - (b) Variant and Time-variant systems
 - (c) Stable and unstable systems

Module -2

- 13 a) Find the exponential Fourier Series for the signal shown in Fig.1. (9)
- Given $T = 2\pi$ and $V = 10$.

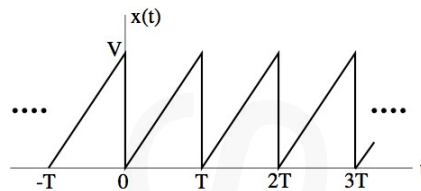


Fig.1

- b) State and prove the time shifting property of Fourier Transform. (5)
- 14 a) Derive the transfer function $\frac{X(s)}{F(s)}$ of the mechanical system shown in Fig.2 (9)

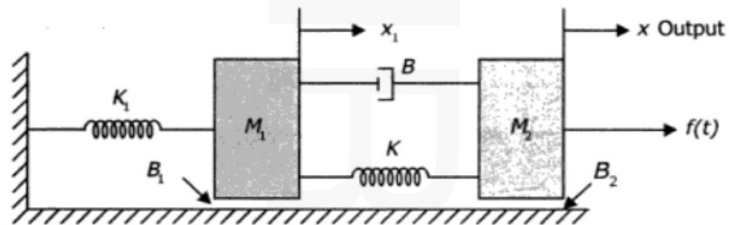


Fig.2

- b) A certain system is described by the following differential equation (5)

$$\frac{d^2 y(t)}{dt^2} + 7 \frac{d y(t)}{dt} + 12 y(t) = x(t)$$

Use Laplace transform to determine the response of the system to unit step input applied at $t = 0$.

Assume the initial conditions on the system are $y(0) = -2$, $\frac{dy(0)}{dt} = 0$

Module -3

- 15 a) Determine the overall transfer function using block diagram reduction for system (5) shown in Fig.3.

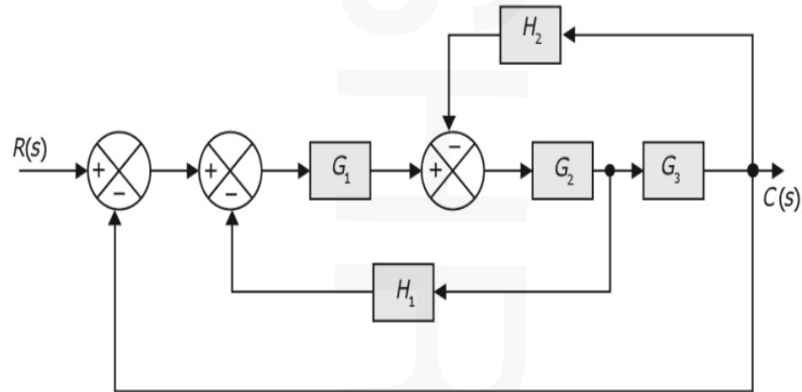


Fig.3

- b) For the closed loop system shown in shown in Fig.4, find the transfer function (9) and thus check the stability of the system if $G(s) = \frac{s}{s+2}$ and $H(s) = \frac{s}{s+1}$ using Routh Hurwitz criterion.

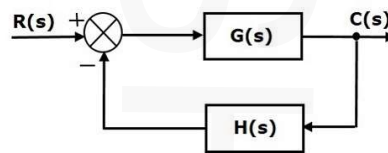


Fig.4

- 16 a) Determine the transfer function of the system shown in Fig.5 represented by the (9) signal flow graph using Mason's Gain Formula.

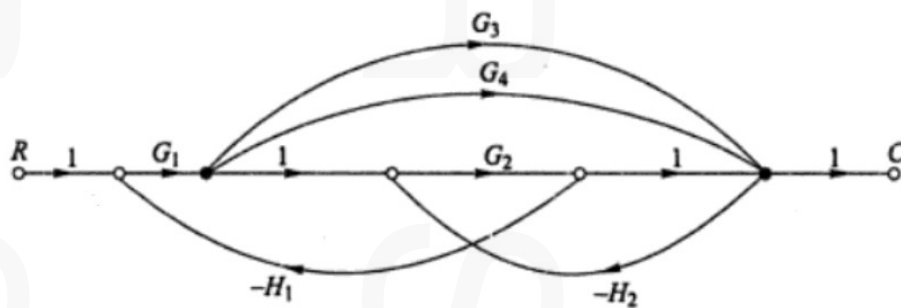


Fig.5

- b) Check the stability of the system represented by the following characteristic (5) equation using Routh stability criterion: $s^3 + 3s^2 + 4s + 2$

Module -4

- 17 a) Determine the convolution sum of the 2 sequences (5)

$$x(n) = \{1, 2, 3, 4\} \text{ and } h(n) = \{1, 2, 1\}$$

- b) Show that the sequences $x(n) = a^n u(n)$ and $y(n) = -a^n u(-n - 1)$ have same Z-transform but different ROC. (9)

- 18 a) Explain the aliasing effect in sampled data systems. (5)

- b) Find the inverse of z-transform of: (9)

$$H(z) = \frac{3z^{-1}}{(1-z^{-1})(1-2z^{-1})}. \text{ Given ROC is } |z| > 2$$

Module -5

- 19 a) A causal discrete time LTI system is described by (9)

$$y(n) - \frac{3}{4}y(n-1) + \frac{1}{8}y(n-2) = x(n)$$

where $x(n)$ and $y(n)$ are the input and output of the system, respectively.

Determine the transfer function $H(z)$ of the system. Also find the step response of the system.

- b) Determine the Fourier series of the sequence $x(n) = \cos 0.2\pi n$ (5)

- 20 a) Obtain the Direct form-I realization for the system described by the difference equation (9)

$$y(n) + \frac{1}{4}y(n-1) - \frac{1}{8}y(n-2) = x(n) - \frac{7}{4}x(n-1) - \frac{1}{2}x(n-2)$$

- b) A system is described by the following characteristic equation: $5z^4 + 2z^3 + Kz^2 + 3z + 1$ Using Jury's test, find the value of K that will result in a stable system. (5)
